BDA LAB04

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Batch: A4

Aim: To learn about multi-cluster setup using hadoop and achieve it.

Cluster of 5 computers

core-site.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<?xml-stylesheet type="text/xsl" href="configuration.xsl"?>
<!--
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  limitations under the License. See accompanying LICENSE file.
-->

<!-- Put site-specific property overrides in this file. -->

<configuration>
<property>
        <name>fs.defaultFS</name>
        <value>hdfs://192.168.29.18:9000/</value>
  </property>
  <property>
        <name>hadoop.tmp.dir</name>
        <value>/opt/hadoop/tmp</value>
  </property>
</configuration>
```

hdfs-site.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<?xml-stylesheet type="text/xsl" href="configuration.xsl"?>
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  WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
  See the License for the specific language governing permissions and
  limitations under the License. See accompanying LICENSE file.
-->

<!-- Put site-specific property overrides in this file. -->

<configuration>
  <property>
          <name>dfs.namenode.name.dir</name>
          <value>/opt/hadoop/dfs/name</value>
  </property>
 <property>
          <name>dfs.datanode.data.dir</name>
          <value>/opt/hadoop/dfs/data</value>
  </property>

<property>
          <name>dfs.hosts</name>
          <value>/opt/hadoop/etc/hadoop/slaves</value>
  </property>
  <property>
          <name>dfs.namenode.http-address</name>
```

```
        <value>192.168.29.18:50070</value>

  </property>
  <property>
        <name>dfs.namenode.secondary.http-address</name>
   <value>192.168.29.18:50090</value>
  </property>
</configuration>
```

yarn-site.xml

```
<?xml version="1.0"?>
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  WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
  See the License for the specific language governing permissions and
  limitations under the License. See accompanying LICENSE file.
-->
<configuration>
<property>
   <name>yarn.resourcemanager.hostname</name>
   <value>192.168.29.18</value>
</property>
<property>
   <name>yarn.nodemanager.aux-services</name>
   <value>mapreduce_shuffle</value>
</property>
</configuration>
```

mapred-site.xml

```
<?xml version="1.0"?>
<?xml-stylesheet type="text/xsl" href="configuration.xsl"?>SS
<configuration>
  <property>
        <name>mapreduce.framework.name</name>
        <value>yarn</value>
  </property>

  <property>
        <name>mapreduce.jobhistory.address</name>
        <value>192.168.29.18:10020</value>
  </property>

  <property>
        <name>mapreduce.jobhistory.webapp.address</name>
        <value>192.168.29.18:19888</value>
  </property>
<property>
  <name>yarn.app.mapreduce.am.env</name>
  <value>HADOOP_MAPRED_HOME=${HADOOP_HOME}</value>
</property>
<property>
  <name>mapreduce.map.env</name>
  <value>HADOOP_MAPRED_HOME=${HADOOP_HOME}</value>
</property>
<property>
  <name>mapreduce.reduce.env</name>
  <value>HADOOP_MAPRED_HOME=${HADOOP_HOME}</value>
</property>

</configuration>
```

slaves

```
hadoop-clone-16
hadoop-clone-17
hadoop-clone-18
hadoop-clone-19
```

hadoop-clone-20

To copy on all nodes in cluster, genslaves.sh

```
labsubnet=192.168.29
sampleconfd=/opt/hadoop/etc/hadoop
for i in {16..20};
do
        rsync -rv $sampleconfd/ $labsubnet.$i:/opt/hadoop/etc/hadoop/
done;
```

For cleanup on all nodes, 6cleanup.sh

```
labsubnet=192.168.29
for i in {16..20};
do
  ssh $labsubnet.$i "rm -rf /opt/hadoop/logs/* /opt/hadoop/tmp/* /opt/hadoop/dfs/*"
done;
```

Format on Namenode

```
hadoop namenode -format
```

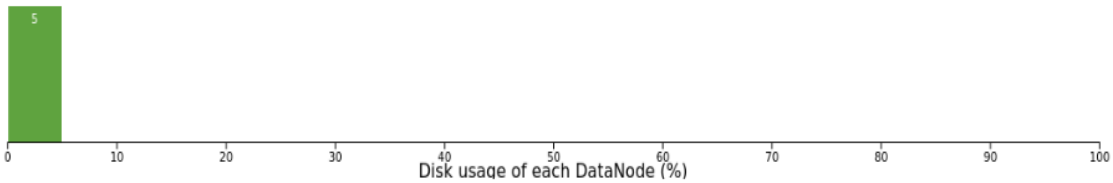
To start on all nodes, start-all-dfs.sh

```
labsubnet=192.168.29
for i in {16..20};
do
        ssh $labsubnet.$i "/opt/hadoop/sbin/start-dfs.sh"
done;
```

Result:

✔ In service ✘ Down ⊘ Decommissioning ⊘ Decommissioned ⊘ Decommissioned & dead
Entering Maintenance In Maintenance In Maintenance & dead

Datanode usage histogram



In operation

DataNode State All Show 25 entries Search:

Node	Http Address	Last contact	Last Block Report	Used	Non DFS Used	Capacity	Blocks	Block pool used	Block pool usage StdDev	Version
✔/default-rack/hadoop-clone-20:9866 (192.168.29.20:9866)	http://hadoop-clone-20:9864	1s	0m	24 KB	50.58 GB	454.9 GB	0	24 KB (0%)	0%	3.4.0
✔/default-rack/hadoop-clone-16:9866 (192.168.29.16:9866)	http://hadoop-clone-16:9864	2s	1m	24 KB	41.87 GB	454.9 GB	0	24 KB (0%)	0%	3.4.0
✔/default-rack/hadoop-clone-17:9866 (192.168.29.17:9866)	http://hadoop-clone-17:9864	1s	1m	24 KB	62.58 GB	454.9 GB	0	24 KB (0%)	0%	3.4.0
✔/default-rack/hadoop-clone-18:9866 (192.168.29.18:9866)	http://hadoop-clone-18:9864	2s	0m	24 KB	60 GB	454.9 GB	0	24 KB (0%)	0%	3.4.0